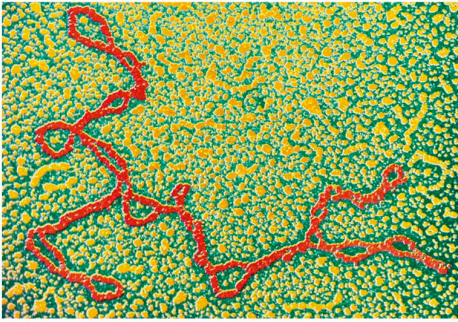




MITOCHONDRIAL DNA

This electron microscope image shows that mitochondrial DNA forms a closed loop, unlike the DNA in the nucleus of the cell, which is linear.

MITOCHONDRIAL GENES

Mitochondria, the powerhouses of the cell, have their own DNA. Unlike DNA in the nucleus, mitochondrial DNA (mtDNA) is circular, not linear. It contains just 37 genes which code for the proteins and RNA the mitochondrion needs for its functions. Mitochondrial DNA is unique in being inherited only from the maternal line, via the mitochondria present in the egg at fertilization. This type of DNA has been used to study genetic relationships and reunite families, as the high rate of mutation of mtDNA means unrelated people have very different mtDNA. Certain rare diseases are associated with changes in the mtDNA.

GENETIC CONTROL OF CELLS

Not all genes are active in all cells. The process by which a gene is able to make its protein is called gene expression. The expression of each gene is regulated, or controlled, according to exposure to chemicals such as growth factors and regulators – products of other genes. Some genes are “switched on” and express themselves in most cells. These are concerned with basic processes such as utilizing glucose for energy. Other genes are “switched off” unless they are needed; these are for making specialized products, such as hormones. As cells’ genes are switched on and off in different circumstances, they differentiate, or become different.

CELLULAR DIFFERENTIATION

The first cells produced by divisions of a fertilized egg are “generalized” stem cells. As they multiply, pre-programmed instructions begin to act. Intercellular contacts and the chemical environment inform cells in certain parts of an embryo to differentiate into tissues such as nerves, muscle, and skin.

PRECURSOR CELL

This can become any of a variety of cells. Some lines of offspring cells retain the ability to generalize, while others go on to become specialists

PRECURSOR CELL

